

```
$misc/unix_like/linux/QR; m4ps 0 1
```

commands here are generic, except for those under the Ubuntu heading, see also \$OSAB/QR/QR.md

```
dotnet --list-runtimes
```

```
dotnet --list-sdks
```

```
info info
```

```
wcsf=$(wc -l <samplefile>); echo $wcsf
```

```
$cITcr/unix-like/usr_lib_X11_rgb.txt # colours
```

```
$misc/GRs/DCGRs.sh
```

```
$OSL/bashrc-generic
```

Pipe Viewer

audio - ALSA

```
alsamixer -V all
```

```
speaker-test -c 2
```

```
sudo alsactl store
```

audio - cmus

```
:Man cmus
```

```
cmus --help
```

```
cmus --plugins
```

```
cmus_notify -h
```

- C* Music Player
- sends low urgency notifications

commands

```
7 settings
```

```
^c echo Type :quit<enter> to exit cmus.
```

```
^l refresh
```

```
space win-toggle => mark
```

```
e win-add-q
```

```
P win-mv-before
```

```
p win-mv-after
```

```
b player-next
```

```
x player-play (from start)
```

```
v player-stop
```

```
z player-prev
```

```
h, left seek -5s
```

```
l, right seek +5s
```

```
, seek -1m
```

```
. seek +1m
```

```
^r toggle repeat_current
```

```
C toggle continue
```

```
r toggle repeat
```

```
-          vol -10%
+ =        vol +10%
```

media

- can play: mka, ogg, opus
- can't play: m4a, oma, omv, rmj, wma

audio - convert

```
f=sox.flac; ffi $f -c:a libvorbis -aq 4 "${f%.*}.ogg"
for f in *.flac; do ffi "$f" -c:a libvorbis -aq 4 "${f%.*}.ogg" ; done
for f in *.oma; do ffi "$f" -c:a libvorbis "${f%.*}.ogg" ; done # at default VBR quality 3
for f in *.rmj; do ffi "$f" -c:a libvorbis -aq 4 "${f%.*}.ogg" ; done
for f in *.wav; do ffi "$f" -c:a libvorbis -aq 4 "${f%.*}.ogg"; rm "$f"; done
for f in *; do ffi "$f" -b:a 128K -vn "${f%.*}.mp3" ; done
```

m4a

```
for f in *.m4a; do ffi "$f" -c:a libvorbis -aq 4 "${f%.*}.ogg" ; done
```

VBR quality 4 is closer to the original size

audio - gst123

```
gst123 -Z . # play random audio files recursively forever
```

control

```
left/right -> seek 10 seconds
down/up -> seek 1 minute
PgDn/Up -> seek 10 minute
space -> toggle pause
+/- -> change volume by 10%
m -> toggle mute
n -> next file
q -> quit
? -> help
```

audio - MPD

```
mpd
mpd --kill
pgrep mpd
```

- can play: m4a, mka, ogg, opus, wma
- Music Player Daemon

vimpc - normal mode

- -/+ decrease/increase volume
- ZZ quits completely
- space start/stop
- backspace/s stop playlist

song

- f/F scroll to current song
- e toggle info
- E toggle repeat
- I restart the song
- R toggle random
- S toggle single
- </> previous/next song

audio - playerctl

```
playerctl # quick guide
playerctl -l # (--list-all) available players - firefox, mpd
playerctl metadata # from the current player
playerctl pause
playerctl play
playerctl play-pause # toggles
playerctl status
playerctl stop
```

audio - PulseAudio

```
pactl -h
pactl info
pactl list short
pgrep -af pulseaudio
pulseaudio -k # --kill
pulsemixer
```

libpulse

```
pacat --list-file-formats
pactl list sinks short
pactl list sources short
pactl set-sink-mute 0 toggle
pactl set-sink-volume 0 60% # master volume to reasonable level
parecord -d 0 parecord.flac
parecord -d 1 parecord.flac
parecord -d 16 parecord.flac
pavucontrol
```

audio - SoX

```
rec -c 2 sox.flac # record in stereo
soxi <audioFile> # info, including duration
```

bc

```
bc -q
bc <<< 'scale=2; 3 * 2.004 / 1' # the final redundant division fixes scale
```

- "an arbitrary precision calculator language"
- command: scale=n => results to n decimal places

limitations

- <circumflex>i = to power of integer i
- can be inaccurate

CopyQ

```
copyq tab # lists the tab headings
copyq tab clipboard read 0 # pastes the first one
pkill copyq; copyq &
```

internal commands

- Ctrl+Home = move_to_top
- Ctrl+p = Preferences

datetime

strftime

cal

```
cal " this month
cal (command)
cal -3 " last month, this, next
cal -n 18 " all 18 months from now
cal -w[y] " week numbers
cal -y " year
```

date

```
date '+%F %a %H:%M'
date +%j-%H%M
date +%s; sleep 1; date +%s
date +%Y%m%d
date +%y%m%d-%H%M
date +%y%m%d-%H%M%S
date -d 'now -1 year'
date -d @<unixTimeToConvert>
date -R
date -r <fileToGetDateOf>
```

- %j day of year (001..366)
- date -d @\$((1767468248734330/1000000)) microseconds, as in sqlite3's .dump

documenting

aspell dicts

Foliate

- ctrl-q = quit
- mouse to bottom for scrollbar
- mouse to top for adjustments

PDFs

- Firefox shows Document Outline
- Outline = the hyperref bookmarks that correspond to headings of a LaTeX document

C5 printing

- Firefox can't figure it's portrait
- LibreOffice Writer gets it right
- Ubuntu's qpdfview gets it almost right

qpdfview

F6 = View > Docks > Outline (such as the headings of a LaTeX document)

F8 = Thumbnails

sizes

pdfinfo <a_pdf_file> # A5 = 420x595. See GSview for more sizes.

1 PostScript point = 0.3528 mm

XpdfReader

[-f <firstPage>] [-l <lastPage>]

pdftoppm -png -r 300 <pdf> <basename_for_png_sequence>

pdftoppm -jpeg -r 300 <pdf> <basename_for_jpeg_sequence>

Zathura

zathura --mode=fullscreen a.pdf &

zathura man page

commands

:info

- <tab> toggles Outline view
- f11 toggle fullscreen
- r rotate by 90 degrees
- R reload document

resizing

- +/-/= zoom in/out/original
- a best-fit mode
- s width mode

Emacs

ctrl-x > ctrl-c => quit

emacs -mm & # maximized

emacs -nw # in terminal

emacs -nw --debug-init

view movements

alt-v => scroll up

ctrl-l => centre text in view

ctrl-v => scroll down

encoding

```
delta <file1> <file2>
git diff <file1> <file2>
~\.pyenv
```

Make (software)

Git

```
:Man git
fd -tf -u index.lock -x rm
git extras --version
gitui
```

Git - configurations

```
$ fd -HI -tf ^config$ | xargs rg -l 'remote = gh' # ripgrep
$ find . -wholename '*.git'
$ find . -wholename '*.git/config' > gitconfig-all.txt
$ find . -wholename '*.gitignore' > gitignore-all.txt
$ find . -wholename '*.git/config' | wc -l
$ grep -rl --include "config" harriott . > gitconfig-harriott.txt
```

gitconfig

```
r ~/.ssh
```

```
$machBld/jo/gitconfig includes $misc/CP/gitconfig
```

TeX

```
pdfjam --help
```

TeX Live

```
/etc/texmf/web2c/fmtutil.cnf
sudo tlmgr -gui
sudo tlmgr update -all -dry-run
```

tests

```
pdflatex sample2e.tex
tex --version
```

texmf.cnf

```
$ABjo/texmf.cnf
```

defines variables for kpathsea

Fcitx Clipboard

- fcitx5-configtool & > Addons > Clipboard > Configure > Trigger Key defaults to ctrl+;
- works in Firefox, WezTerm

fcron

```
fcrontab -l
sudo systemctl restart fcron.service
systemctl status fcron.service
```

file contents

```
cat
diff <file1> <file2>
enca -l surfaces
printf "line terminated with CRLF\r\n" > crlf.txt
tac
shuf
sk --ansi -i -c 'rg --color=always --line-number "{}"'
sort -ro <file> <file> # reverse sort in place
wc -l <file> # counts lines
xargs < <file_to_return_as_one_line>
```

- "CRLF line terminators"
- sharkdp/bat
- `uniq -c` (`--count`) prefix lines by counts

awk

```
awk '{print $0}' # prints all columns - doesn't work with OFS
awk '{print $1}' ORS='\t' <file_with_column_to_print_tsv>
awk -f script.awk input.txt
awk -F, '{print sprintf("%02d",NR),NF}' file_to_check_field_counts.csv
awk -h # --help
awk -i inplace -F, '{print $3,$2,$1}' OFS='|' toReorder.csv
awk -V # --version
awk 1 ORS=' ' <file_to_return_as_one_line>
v=variable; awk -v var="$v" 'BEGIN {print var}'
za $cIThul/gawk.pdf
{print $1,sprintf("%08d",$2)} # 2nd field is 0-padded to length 8
```

- `'{if($1==2){if(gsub(/|/, "|")<1){$3=$3"|"} } {print $1,$2,$3}}'` adds a `|` if there aren't 2
- `-F` = `--field-separator`
- `-i` = `--include`
- GNU Awk
- `gsub(/old/, "new" [, target])` returns the number of substitutions made
- no backreferences
- `substr($4,2,length($4)-2)` removes outer characters

built-in variables

- `FILENAME` name of the current input-file
- `FS` input field separator character (default = any space and tab characters)
- `NR` number of input records seen so far
- `NF` number of fields in an input record
- `OFMT` format for numeric output (default = `%.6g`)
- `OFS` output field separator (default =)
- `ORS` output record separator (default = `\n`)
- `RS` record separator (default = newline)

octal codes

- `\047` = `'`
- `\057` = `/`

input field separation

- F,
- F'\t'

csv

- F,
- -vFPAT='([^\,]*)|("[^"]+")' f1,"f2a, f2b",f3

numbers

```
awk -F, '{ if ( $1+0 > 2025 ) print $1 }' <first_column_could_be_year.csv> # only those > 2025
awk '{ print ($1 % 2 == 0) ? "even" : "odd" }' numbers.txt
awk '{printf("%06.2f\n", $1)}' <<< -9.111111
o -e "a 1\nb 2\nc 3" | awk '{s+=$2;}END{print s;}' # sum of column 2
o -e "a 1\nb 2\nc 3" | awk '{print $2}' ORS=' ' | sed 's/ $/\n/'

$1 ~ /^[0-9]+$ / checks if first field is an integer
```

select lines

```
awk 'FNR==2 {print}' <file> # line 2
awk '{if(NR>=2){print}}' <file> # lines from 2
```

dos2unix

```
dos2unix -i *
fd -tf -x dos2unix -ic # list files that would be converted

-e (==--add-eol) not on WSL Ubuntu
```

file

```
fd -Itf -x file
```

Vim fileencoding utf8 reported as ASCII

grepping

```
grep -c '^PatternAtStartOfLine' <file> # returns count of occurrences
grep -P '[\p{Devanagari}]' **/*.md # finds Devanagari characters
if grep -q '†' <file_that_may_contain_†>; then o †; fi
```

- -A num (--after-context=num)
- -i (--ignore-case)
- -E (--extended-regexp)
- -F (interpret patterns as --fixed-strings, not regex)
- -o (--only-matching)
- -q (--quiet, --silent) no standard out
- GNU Grep Manual

igrep

```
ig -h # --help
```

commands

- ? toggle help
- f5 toggle search pattern
- <enter> edit the file on the line
- <esc>/q quit

context view

- s toggle horizontal
- v toggle vertical
- -/+ decrease/increase size

sed

\$cIThul/sed

:Man sed

echo "blia blib bou blf" | sed -E 's/bl(ia|f)//g'

sed --version

sed 5q <file> prints first 5 lines

sed G <file> # double-spaced

- -E/-r (--regexp-extended) extended regular expressions
- GNU sed
- linewise, so don't try to replace newline
- stream editor

bracket expressions

echo ' a' | sed 's/^[[:space:]]//'

echo '20230516' | sed -r 's/(20[0-9][0-9])([0-9][0-9])/\1-\2-/' # 2023-05-16

echo 'gray, grey' | sed 's/gr[ae]y/blue/g'

character classes

- [:alpha:] = [:lower:] + [:upper:] = [A-Za-z]
- [:alnum:] alphanumerics ([:alpha:] + [:digit:] = [0-9A-Za-z])
- [:digit:]: could echo '12' | sed 's/[[[:digit:]]//', but echo '12' | sed 's/[0-9]//' is more compact

make changes

echo -e '1\n2\n3' | sed 's/.*\t&/g' # inserting tabs

echo "don't forget that" | sed 's/\x27/\\"/'

echo "é" | sed 's/é/e/'

echo "0o" | sed 's/o/a/Ig' # case insensitive

echo "THIS is a test!" | sed 's/.*\L&; s/[a-z]*\u&/g' # title case

sed -i 's/\t/tab_gone/g' <file_with_tabs>

sed -i '/<regex>/!d' <file_to_reduce> # removes lines that don't match

sed -i '/match/,+2d' <file> # removes matched line and 2 after

sed -i '0~2 a\\' <file_to_add_blank_line_after_each_2nd_line>

sed -i 's/\r//' <file_to_remove_CRLF_from>

sed -e '0,/first/ s/first/this_first_only/' -i <file_to_make_just_one_single_change_to>

- & whole matched pattern
- replace a string in multiple files

delete lines

sed -i '1,4d' <file>

sed -i '2d' <file>

sed -i '\$d' <file_to_lose_last_line>

prepend

sed '1i\nNewFirstLineText' <file_to_prepend_to>

sed -i '1s/^/vim: ft=<filetype>:\n\n/' <file_to_prepend_vim_modeline>

show file contents

```
sed '/pattern/q' <file> # cat's <file> until pattern
sed 5q <file> print first 5 lines
sed -e '' <file> # mimics cat
sed -n '/<regex>/p' <file> # prints only the matching lines
sed -n '2,$p' <file> # prints from the 2nd line
```

file manage

```
$cGRs/unix/ranger-ranger
fuseiso <ISO_image> <mountDirectory>
mkdir -p # --parents = make parent directories as needed (no error if existing)
sudo chown -R <user>:<group> <dir> # --recursive
```

- \$OSL/nodes/bashrc-console-fm nnn
- chmod 600 file - owner can read and write
- chmod 644 file - owner can change it, everyone else can read it
- chmod 660 file - owner can read and write, and group members
- chmod 666 file - all can read and write
- chmod 700 file - owner can read, write and execute
- chmod 711 file - drwx--x--x
- chmod 755 file - drwxr-xr-x
- chmod 777 file - all can read, write and execute
- FILE(1)
- fsearch: Ctrl+p = Preferences
- install(1)
- rm(1)

compressed

:Man gzip

gzip -r * will recursively convert all files to *.gz

7-Zip

```
7z -h # -mhe (metadata encryption) not there
7z.exe a -mhe -p<pw> dir.zip dir
for z in *.zip; do 7z x $z; done
```

compression level

- -mx0 less than no option
- -mx1 seems optimal

tar

:Man tar

```
tar -xzf archive.tar.gz [-C <target_directory>]
```

- --gunzip (or --ungzip)
- -v (--verbose)
- -z (--gzip)

zip

:Man zip

```
zip -qr dir.zip dir
```

- -q (--quiet)
- -r (--recurse-paths)

cp

`cp -r <sourceDir> .`

`-L, --dereference`

digests

`md5sum -c <md5file>`

`rhash -a --bsd <somefile>`

`rhash -h # --help`

`rhash -M # --md5`

`rhash --list-hashes => list names of all supported hashes`

`rhash --sha256 <file>`

`rhash --sha256 * # to see if files are identical`

`rhash --sha512 <file>`

`shasum -a 256 <file>`

Dolphin

`F4` -> Konsole attached below

`Shift+F4` -> Konsole in a new window

fd

`fd -tl -HL -X rm # removes dead links`

`fd -tf stderr.txt -X rm`

find

`find $PWD -name <file> # gets full path`

`find -cmin 1 # files whos status changed in last minute`

`find -regex ".*a.*\|.*b.*"`

`find . ! -wholename '*/.git/*' -printf %TY%Mm%Td-%TH:%TM:%.2TS\ %p\\n | sort # excluding co`

`find . -maxdepth 1 -mindepth 1 -type f -name "*" # those in working directory`

`find . -name "*" -type f ! -path '*/.git/*'`

`find . -name '*.txt' ! -name 'build*' # excluding build*`

`find . -newer oldFile`

`find . -path '*exclude_path*' -prune -o -name 'partial_filename*' -print`

`find . -type f -exec du -h {} + | sort -hr > descendingSizes.txt`

`find . -xtype l -delete # quickly removes broken symlinks`

`find(1)`

GIO

`gio help`

`gio version`

(Gnome Input/Output)

investigations

`diff --no-dereference -qr dir1 dir2`

`find . -name '.*' # recursively list hidden files`

`for n in *; do find "$n" | echo "$n"; done`

```
stat -c '%a %n' * # show octal permissions
```

by extension

```
find . -name '*.*' -type f | rev | cut -d. -f1 | rev | tr '[:upper:]' '[:lower:]' | sort | un
-count | sort -rn > ext_counts
find . -type f -name "*.*" | grep -o -E "\.[^\.]+\." | grep -o -E "[[:alpha:]]{2,16}" | awk '{p
u
find . -type f | grep -oE '\.(\w+)\.' | sort -u
find . -type f | sed -rn 's|.*[/^\.]+\.[^\.]+\.$|\1|p' | sort -u # extensions list (no hidden
for f in **/*.*; do echo "${f##*}"; done | sort -u
```

counts

```
echo `find . -type d | wc -l`-1 | bc # counts all subdirectories
find . -name "*" -type f -path '*/.git/*' | wc
for d in $(fd -dl -td); do find "$d" | o "$(wc -l) : $d"; done # files in directories
isutf8 **/* | wc -l # non UTF-8 files (fails when too many)
```

including hidden

```
find . | wc -l # very fast in $Drpbx
find . -type f | sed 's/\.\./.*\./' | sort | uniq -c # by extension
```

mlocate

```
locate .asc | grep '\.asc$'
locate .dat | grep '/\.dat'
locate -c <target> # --count
locate -h
```

most recent

```
find . -type f -printf '%TY-%Tm-%Td-%TH:%TM:%.2TS %p\n' | sort -k1,1r | head -5
ls -At | head -n5
```

sizes

```
diskonaut -h
dust -x / # graphical size representation of root, not descending into mounts
```

listed

```
ls -lRhs | sed -e "s/^ *//" | grep "[0-9]" | sort -hr | head -n50 # neat recursive list of la
```

du

```
du -ah . | grep -v "/$" | sort -rh | head -44 # recursive list of largest 44 files
du -h --max-depth=0 <directory>
du -h --max-depth=1 # for a quick list of folder sizes
du -hs # size of current directory
du -hs <glob>
du -s **/.git
```

- -c = grand --total of list
- -h = --human-readable
- -s ---summarize, don't recurse
- -x = --one-file-system
- du(1)

ncdu

```
ncdu [<directory>]
```

commands

- ? -> help
- g -> rotate through % graph none
- n -> order by name
- s -> order by size

symlinks

```
[[ -L $s ]] && echo 'symlink exists'
```

```
find . -type l -ls # recursively list all symlinks with their references
```

```
readlink $s # returns referenced node
```

tree

```
tree -a # include hidden
```

```
tree -d [-L n] # directories only, maximum Level (= depth) n
```

```
tree -f # print path
```

```
tree -fi # print path without indentation lines
```

- -l follows symbolic links, but still showing -> <real_path>
- tree(1)

ln

- -s (--symbolic) not hard
- ln(1)

ls

```
dircolors # LS_COLORS=...
```

```
dircolors --help
```

```
dircolors --version
```

```
dircolors -p # --print-database
```

```
ls -l # show permissions, owners, exact sizes, and date-time
```

- -d (= --directory)
- ls(1)

rm with Pipe Viewer

1. find <directory> | wc -l # gets the <filecount>
2. sudo rm -rv <directory> | pv -l -s <filecount> > /dev/null # shows progress for rm

rsnapshot

```
ncdu --exclude *rsnapshot
```

```
rsnapshot configtest
```

```
sudo rsnapshot du # can takes many hours...
```

```
systemctl list-unit-files | grep rsnapshot
```

```
/usr/bin/rsnapshot sync: completed successfully
```

rsync

```
rsync -irtv --delete $TeNo/ $Drpbx/Play1/TextNotes 2>&1 | tee $Drpbx/Play1/TextNotes_rsync.
```

```
rsync -inrtv --delete --progress path1/large_file_dir1/ path2/large_file_dir2
```

options

```
--exclude=PATTERN
```

- `-h (--help)`
- `-i (--itemize-changes)`
- `-l (--links)` copy symlinks as symlinks (so no "skipping non-regular file" if the symlink's reference exists)
- `-L (--copy-links)` transform symlink into referent file/dir
- `-n (--dry-run)`
- `-r (--recursive)`
- `-t (--times)` keep modification times
- `-v (--verbose)`
- `-V (--version)`

for system files

- `-A` (`--acls`) keep ACLs, implies `-p`
- `-a` (`--archive = -Dgloprt, no -A, -H, -X`) handles symlinks with no reference
 - `-D` (`--devices --specials`)
 - * `--devices` keep device files
 - * `--specials` keep special files
 - `-g` (`--group`)
 - `-o` (`--owner`) if super-user
 - `-p` (`--perms`) keep permissions
- `-H` (`--hard-links`)
- `-X` (`--xattrs`) keep extended attributes

output info

YXcstpoguax path/to/file

```
| | | | | | | | | |  
| | | | | | | | | | x: The extended attribute information changed  
| | | | | | | | | | a: The ACL information changed  
| | | | | | | | | | u: The u slot is reserved for future use  
| | | | | | | | | | g: Group is different  
| | | | | | | | | | o: Owner is different  
| | | | | | | | | | p: Permission are different  
| | | | | | | | | | t: Modification time is different  
| | | | | | | | | | s: Size is different  
| | | | | | | | | | c: Different checksum (for regular files), or  
| | | | | | | | | |     changed value (for symlinks, devices, and special files)  
| | | | | | | | | | the file type:  
| | | | | | | | | | f: for a file,  
| | | | | | | | | | d: for a directory,  
| | | | | | | | | | L: for a symlink,  
| | | | | | | | | | D: for a device,  
| | | | | | | | | | S: for a special file (e.g. named sockets and fifos)  
| | | | | | | | | | the type of update being done::  
| | | | | | | | | | <: file is being transferred to the remote host (sent)  
| | | | | | | | | | >: file is being transferred to the local host (received)  
| | | | | | | | | | c: local change/creation for the item, such as:  
| | | | | | | | | |   - the creation of a directory  
| | | | | | | | | |   - the changing of a symlink,  
| | | | | | | | | |   - etc.  
| | | | | | | | | | h: the item is a hard link to another item (requires --hard-links).
```

- .: the item is not being updated (though it might have attributes that are being modified)
- *: means that the rest of the itemized-output area contains a message (e.g. "deleting")

touch

- -c (--no-create)
- change file timestamps

zoxide

```
declare -f z
zoxide -h
```

Flatpak

```
du -sh /var/lib/flatpak/.removed
flatpak -h # --help
flatpak --installations # prints paths
flatpak config -h
flatpak config --list
flatpak info -h
flatpak list
flatpak remotes # list added repositories
flatpak history
flatpak install ... # recovers from network outages
flatpak uninstall <pkg>
sudo flatpak list
sudo flatpak repair # Erasing .removed
sudo flatpak update
```

fortune

```
fortune -a
fortune -f
```

get at root on tty2

```
Ctrl+Alt+F2 > root + pw
```

GNU Privacy Guard

```
echo "encrypt me this" | gpg -ase -r jharr
gpg --full-gen-key
```

```
fingerprint: xxxx xxxx xxxx xxxx xxxx  xxxx xxxx xxxx xxxx xxxx
```

keybox

```
gpg --delete-key <anID>
gpg --import my_private_key.asc
gpg --import my_public_key.asc
gpg -ao private.asc --export-secret-keys jh
kbxutil ~/.gnupg/pubring.kbx | moar # for a more detailed exploration
```

hw

```
lsusb  
solaar show  
sudo chmod 777 /run/media/jo/TOSHIBA  
sudo chmod 644 $0bc/autostart/urxvtl.sh  
uname -m # reports eg x86_64  
xbacklight -set 50  
:e /run/media/jo
```

battery

```
acpi -h  
batstat
```

CPU

```
cpupower frequency-info  
sudo cpupower frequency-set -u 3400Mhz  
sudo turbostat --interval 0.1 --num_iterations 1  
sudo turbostat --interval 2 --quiet
```

graphics

```
cat /sys/class/graphics/*/virtual_size  
xdpyinfo | grep dimensions: | awk '{print $2}' # combined screen resolution  
xrandr # all available monitor resolutions  
xrandr | grep 'connected'  
xvidtune -show # Modeline-style information
```

fbset

```
fbset -h  
sudo fbset
```

framebuffer device settings

query the card specs

```
echo $(xrandr -q | grep '*' | uniq | awk '{print $1}' | cut -d 'x' -f1) $(xrandr -  
q | grep '*' | uniq | awk '{print $1}' | cut -d 'x' -f2) # screen size  
GPU=$(lspci | grep VGA | cut -d ":" -f3);RAM=$(cardid=$(lspci | grep VGA | cut -d " " -  
f1);lspci -v -s $cardid | grep "prefetchable" | cut -d "=" -f2);echo $GPU $RAM  
lspci | grep -e VGA -e 3D # gets just the graphics card name
```

keyboard

```
xev | awk -F'[ ]+' '/^KeyPress/ { a[NR+2] } NR in a { printf "%-3s %s\n", $5, $8 }' # scancod
```

for XF86 symbols

```
xmodmap -pke | grep Audio  
xmodmap -pke | grep Brightness
```

printing - CUPS

```
lnav /var/log/cups  
lpadmin  
lpq # show printer queue status
```



```
lpstat -p -d # show status of printers, including which is set as default
lpstat -s # status, including URL
sudo systemctl status cups.service
```

Command-Line Printing and Options <http://localhost:631/help/options.html>

cancel print jobs

```
cancel -a
lprm
```

lp

```
lp -n <number_of_copies> <image>
lp -o scaling=200 <image> # prints divided across 4 pages
```

can't center image on page

prints all images in a directory

```
lp *
lp *.JPG
lp *.png -o ColorModel=KGray # black cartridge only
```

orientation

automatically selects the best orientation for filling the page

specify

```
-o orientation-requested=3 - portrait orientation (no rotation)
-o orientation-requested=4 - landscape orientation (90 degrees)
-o orientation-requested=5 - reverse landscape or seascape orientation (270 degrees)
-o orientation-requested=6 - reverse portrait or upside-down orientation (180 degrees)
```

printing - HPLIP

```
hp-info # reports version and pops up info
hp-levels # takes a little while to show the ink levels (after "Using device : ...")
hp-printsettings # dialog pop-up, but no dpi's
hp-setup -h
hp-setup -r # GUI to remove printer
hp-systray & disown # HPLIP Status Service tray icon (access to HP Device Manager)
hp-testpage
hp-toolbox & # HP Device Manager
```

webcam

```
cameractrls
```

Guvview

```
gview &
~/my_video-n.mkv
```

Logitech B525: Audio Controls > Audio API > PULSEAUDIO

V4l-utils

```
qv4l2 &
v4l2-ctl --list-devices
v4l2-ctl -d /dev/video0 --list-ctrls
v4l2-ctl -d /dev/video1 --list-ctrls
```

imagey

```
colorpicker --one-shot --preview
gphoto2 --auto-detect # list detected cameras
gphoto2 -DR # delete all files in all folders
gphoto2 -l # list folders
gphoto2 -L # list files
gphoto2 -P # get all files
gpicview # opens first image in directory (no thumbnails)
exiftool <image>
exiftool -Orientation * # reports
exiv2 <image>
```

feh

```
feh -F # fullscreen
feh -l [*] # [recursively] list image specifications
```

- can open webp
- can't animate a gif

in-image

```
d => toggle_filenames
```

size

```
* => 100%
/ => fit the window size
up/down => zoom in/out
```

ImageMagick

```
magick -flatten transparent.png white_background.png
magick -list
magick -list gravity
magick <colourimage> -colorspace Gray <grayimage>
magick <people> -paint 9 <people-oil>
magick <positive> -negate <negative>
for i in $(ls); do magick -resize 10% $i r$i; done
for i in $(ls); do magick -resize 15% $i r$i; done
for i in $(ls); do magick -resize 25% $i r$i; done
for i in $(ls); do magick -resize 50% $i r$i; done
```

convert

```
for i in *.bmp; do magick $i ${i%.*}.jpg; done
for i in *.svg; do magick $i ${i%.*}.ico; done
for i in *.svg; do magick $i ${i%.*}.png; done
for i in *.jpeg; do magick $i jpg/${i%.*}.jpg; done # gets them into subfolder jpg
for i in *.webp; do magick $i ${i%.*}.jpg; done
```

recolour

```
for i in *.png; do magick $i -fill green -colorize 100% ${i%.*}-green.ico; done
for i in *.png; do magick $i -fill orange -colorize 100% ${i%.*}-orange.ico; done
for i in *.png; do magick $i -fill red -colorize 100% ${i%.*}-red.ico; done
for i in *.png; do magick $i -fill yellow -colorize 100% ${i%.*}-yellow.ico; done
```

nomacs

nomacs -h

nomacs -v

- can only fill page for printing
- can't open avif, heic, webp
- PrintPreview-PageSetup-Landscape

keybindings

ctrl+k => keyboard Shortcuts

ctrl+r => resize

ctrl+s => save dialogue

ctrl+shift+p > Close on ESC

f => toggles selected panes

F1 => credits

F10 => frameless

Alt

Alt A > Exposure > Gamma

Alt A > Tiny Planet

not having cropped

Ctrl+w => next tab

Ctrl+w => close tab

Panels

alt+m => Metadata Info

i => File Info

f => hide all

m => Metadata Ribbon

mouse right-click to find them

Pinta

- attempts to shows cursor position in pixels, but underestimates - use GIMP
- backspace (= Edit > Erase Selection)
- can't send to printer...
- Rectangle Select > Cut > Paint Bucket to replace an area with a solid colour fill

pqiv

pqiv -i <animateGif> & # opens the animated gif without the obtrusive info box

pqiv --auto-montage-mode * & # flat recursive view of all imagies

pqiv --show-bindings

- can open avif
- changes aren't saved
- plays mp4's
- only sees specified images
- powerful quick image viewer
- recursive
- shows size in status bar tab

scanimage

scanimage -A => --all-options => lists all available options

scanimage -h

scanimage -L => --list-devices

[n]sxiv

- can't animate a gif
- can't send to printer
- Simple X Image Viewer

commands

thumbnail mode: R reload all

image mode

- n p next previous
- [] 10x backwards forwards
- | _ flip horizontal vertical
- changes aren't saved

pan

- h left
- j down
- k up
- l right

rotate

- < 90° ccw
- > 90° cw
- ? 180°

scroll

- H to left
- J to down
- K to up
- L to right
- z to center

zoom

- = by 100%
- e E fit width height
- F fill (often getting only part of the image)
- w fit or 100%
- w fit (the default)

general

- q quit
- return toggle image/thumbnail
- f toggle fullscreen
- b toggle statusbar
- g G first last
- r reload
- + - zoom in out

nsxiv

```
i nsxiv
nsxiv *
nsxiv <path_to_dir>
nsxiv -c # --clean-cache
tree ~/.cache/nsxiv
```

- can open avif
- only sees specified images

sxiv commands

```
b => toggle infobar
f => toggle fullscreen
q => quit
```

No refresh of directories...

Vimiv

```
vimiv &
```

- can open avif
- can't open a gif
- doesn't pick up changes in the directory

commands

```
O => switch focus
o => toggle the library (the current directory)
w => write changes
```

thumbnails

```
+/- => size
t => toggle
```

tagged file list manipulation

```
:tag_remove tagname
:tag_write tagname
cat ~/.local/share/vimiv/Tags/tagname | while read i; do mv $i .; done
rm -r ~/.local/share/vimiv/Tags/tagname
```

lsof

```
[sudo] lsof -i
```

list open files

multimedia

```
ffprobe -i <avfile> -show_format -v quiet | sed -n 's/duration=//p' # fractional seconds
mediainfo <avfile> | grep Encoded
mediainfo <avfile> | grep Title
```

mpv

```
mpv av://v4l2:/dev/video0
mpv --start=-2 <avfile> # starts 2 seconds from end
```

networking

```
arp-scan -lx # lists subnet hosts
bluetoothctl -- devices
sudo iptraf-ng # ncurses network statistic monitoring utility
sudo lshw -c network
sudo lsof -i -P -n | grep LISTEN # to see the listening ports
```

devices

```
arp-scan --localnet # reports MAC addresses on network
ip link
networkctl list
```

hostname

```
hostname
hostnamectl
uname -n # hostname
```

iproute2

```
ip a # ip address show
ip l # lists machine's ethernet devices
ip neigh # subnet hosts
ip r # ip route show - compactly shows my internal ip address
```

iwd

```
man iwd
sudo ls /var/lib/iwd # reports internet source
systemctl status iwd.service
```

iwctl

```
man iwctl
iwctl device list # shows MAC address
iwctl known-networks list
iwctl station wlan0 get-networks # shows if wlan0 is connected to one
requires a DHCP client to get an IP address
```

NetworkManager

```
nmcli device # list of networking devices
nmcli device wifi list | cat # paged list of SSIDs, with those IN-USE starred
nmcli device wifi connect <SSID> password <pw>
```

saved connections

```
nmcli connection delete <SSID> # can help
nmcli connection show | cat # reports UUIDs and colourizes the active device
nmcli connection up uuid <UUID>
```

NordVPN

```
nordvpn cities United_Kingdom
nordvpn cities United_States
nordvpn countries
```

```
nordvpn rate x # 1 (poor) to 5 (excellent)
nordvpn set technology nordlynx
nordvpn settings
nordvpn status
pgrep nordvpn
sudo kill -9 -f nordvpn # doesn't logout
```

account

```
nordvpn connect CA
nordvpn connect UK
nordvpn connect US
nordvpn disconnect
nordvpn login
nordvpn logout
nordvpn rate 5
nordvpn status | xcol United
```

ss

```
bm ss
ss --tcp --listening
ss -tulw # listening on TCP, UDP, RAW sockets
```

ssh

```
$HOME/.ssh/authorized_keys
ssh localhost
sudo systemctl restart sshd.service
sudo systemctl status sshd.service
```

session commands

~? => supported escape sequences

exit

```
exit
~.
```

npm

which npm

cspell

```
cspell trace "colour" # shows which dictionary it's in
cspell check <filename>
cspell lint --help
```

ffmpeg-concat

- -d <transition_duration> default 500
- -o <out_file> default out.mp4
- -t <transition_name> default fade

npx

```
ls ~/.npm/_npx/*/node_modules
rm -r ~/.npm/_npx/
```

one-line of command output

```
echo $(ls)
ls | xargs
xargs < <file> # cat alternative
```

pass

```
pass # pass ls
pass help
pass init <gpg-id> # creates ~/.password-store/.gpg-id
pass add [-f] <node> > <pw> # insert --force
pass insert <node> > <pw>
pass <node>
pass mv <node_old> <node_new>
pass rm [-f] [-r] <node> # [--force] [--recursive]
pass(1)
```

processes

```
lsof -i
pidof init # process id of init, which is always 1
sudo iotop -o
time sleep 1
xprop # WM_CLASS(STRING) = "instance", "class"
niceness: -20 = highest priority, 19 = lowest
```

kill

```
kill -9 **<tab> # fzf
pkill zoom
```

list

```
ps -ef | grep yt-dlp
ps -ef | grep hp-
ps -efjH
pstree -C age
```

pgrep

```
i pgrep
pgrep <aprogram> # reports it's process id
```

shell

- dc desk calculator (reverse-Polish)
- echo \$SHELL reveals flavour
- fc "fix command"

Bash

```
$misc/linux/QR/script.sh
$culLB/colours
$culLB/Scratch0.sh
```



```

/etc/profile
bash --help
bash --version
bind -P
echo "$floatingpointnumber-$another" | bc # FP math
echo "$PS1"
echo $PWD
echo $SHELL
env | grep SHELL
exit 0 # to quit script
help compgen
kill -9 **<tab> # invokes fzf (if no ble.sh)
printf command
read -p "hit Enter"; echo hello
read a b; echo $a $b
shopt
shopt dotglob # reports back its status
shopt -u nullglob # incase t='${\?}'; echo $t
spectroterm -h
tail -n 1 <file> # last line
za $cITcr/unix-like/linux/GNUOS/bash.pdf
~/.bash_history

```

```

case 'one' in
  one) echo 1;;
  two) echo 2;;
esac

```

- BASH(1)
- command substitution \$(...)

aliases

```

alias
compgen -A alias | awk '{print}' ORS=' : '; echo # compact list
unalias

```

Atuin

```

atuin
atuin stats
atuin stats last friday
atuin stats yesterday
~/.config/atuin/config.toml
~/.local/share/atuin

```

SQLite database of commands

ble.sh

Bash Line Editor

Vim mode

```
ble-bind -P # --print the keybinds
```

- insert-mode: Ctrl-x Ctrl-v = display-shell-version
- normal-mode: K = command-help
- once begun, f1 = command-help

clear screen (saving scrollbar)

```
clear -x # erase the all lines not in scrollbar
ctrl+l => scroll the current line to the top
tput reset
```

conditionals

```
[ "$a" ] && echo $a
[ -z "$a" ] && echo zero_string; [ -n "$a" ] && echo string_not_null
[ true ] && echo y
[ "y" ] && echo y
c=1; [[ $c == 1 ]] && echo "true"
if true; then echo 'not false'; fi
if <condition1>; then <action1>; elif ... else ... fi
if [ "$a" ]; then echo '$a is defined'; fi
t="y"; if [ $t ]; then echo $t; fi
```

=~

```
[[ "hello" =~ "ll" ]] && o ll
v=value; [[ ! $v =~ val ]] && echo val
```

echo

echo "~ stays~ in double quotes, it's unexpanded to \$HOME"

-n no trailing newline

backslash escapes

```
echo -e "\\t" word_after_tab
echo -e '\t' word_after_tab
echo $'aa\bb'
```

-E # (default) no interpretation

file manage

```
echo */ # lists directories
echo <file>
for f in *; do mv $f ${f:2}; done
man -h ls
mktemp temp-XXX # can add more X's, touch's a randomised filename
pushd ~/some_path; pushd /another_path; popd; popd
>f # creates/empties file f
>>f # creates file f if necessary
```

find

```
find . -maxdepth 1 -mindepth 1 -type d -name "*" # those in working directory
find -iname \*.flv -o -iname \*.mp4 -o -iname \*.ogv
find -type f -name "[[:upper:]]*" # files beginning with capital letter

-iregex <pattern> = case insensitive -regex
```

tests

```
[ -d "$d" ] && echo "directory $d is there"
[ -f "$f" ] && echo "file $f is there"
[ -s "nohup.out" ] && echo non-zero file size
[[ -d "$d" && ! -L "$d" ]] && echo "It's a directory and not a symbolic link"
rm a b; touch a; sleep 1; touch b; ls -lt; [ a -ot b ] && echo older
```

functions

```
"$@" = "$1" "$2" ...
```

```
export -f <usefulfunction> # allows it to be used in a script
unset -f <function>
```

lists

```
compgen -A function | awk '{print}' ORS=' : '; echo
declare -F # lists all declare -f possibilities
```

show code

```
declare -f <function_name> # shows its contents
type <function_name>
```

Internal Field Separator - set

```
IFS=this
unset IFS
savedIFS=$IFS; ...; IFS=$savedIFS
```

Internal Field Separator - test

```
cat -et <<<"$IFS"
echo "hello$IFS"."there"
```

job control

```
ctrl+c
bg
fg %n
jobs -l
kill %n
kill -9 %n
```

Login shell?

```
echo $0 # "-bash" = login shell, "bash" = non-login shell
shopt login_shell
```

loops

```
break # breaks out of loop
continue # to next iteration of loop
c=5; for i in $(seq $c); do echo $i; done
for i in {0..9..2}; do echo $i; done
for i in bee fly wasp; do echo $i; done
s=2; e=4; for (( c=$s; c<=$e; c++ )); do echo $c; done
select f in apple pear grape; do echo "you chose $f"; done
while read line; do echo "$line"; done <file_to_use_line_by_line
```

globs

```
$misc/unix_like/linux/need_for_nullglob.sh
for d in */; do echo $d; done
for d in /mnt/*; do echo $d; done
for f in **/*; do echo $f; done
```

managing commands

```
o $? # exit code of last command, 0 if command succeeded
eval $Command
vi mode: v => edit a command in $EDITOR
```

execute last command

```
!!
sudo !!
```

redirecting

```
<somecommand> 2>&1 | tee stdout+stderr.log
<somecommand> > /dev/null 2>&1
```

find ... 2> /dev/null suppresses stderr stream

key

```
1 = stdout file descriptor
2 = stderr file descriptor
>& = redirect to a file descriptor
1>&2 = 2>&1
```

online code check

- <https://explainshell.com/>
- ShellCheck <https://www.shellcheck.net/>

set

```
echo $- # current options
set -e # terminate on a fail
set -v # show code lines as they're used
set -x : show commands and arguments
```

change - to + to remove the option

unset

```
unset **<tab> # invokes fzf (if no ble.sh)
var1=1; var2=2; echo $var1 $var2; unset var1 var2
```

su[do]

```
su - # login shell
```

substitute user identity

trap

```
trap - debug # turn off debug
trap read debug # puts a read request after each executable line
```

variables

```
echo $((2*2))
export # lists environment variables
set # lists all variables
```

arrays

```
a=(1 2 3); ((a[0]++)); ((a[1]+=2)); a+=(4); echo ${a[@]}
a=(1 2 3); echo ${a[-1]} # last element
a=(1 2 3); b=("${a[@]}"); b+=("${a[@]}"); echo ${b[@]}
echo ${#array[@]} # number of elements
firstElement=${array[0]}
string='My string'; [[ $string =~ "My" ]] && echo success
```

don't export them

from file

```
aff=$(<$f); aff=(${aff//'\n'/,}) # array from file $f
mapfile --help
mapfile -t found < <(find .)
mapfile -t array < <(cat file)
mapfile -t GreekArray < <(printf "Alpha\nBeta\nGamma"); echo ${GreekArray[@]}
```

list

```
echo ${array[@]}
for ((index=0; index<${#array[@]}; ++index)); do echo "$index" "${array[index]}"; done
for item in "${array[@]}"; do echo; echo "$item"; done
printf '%s\n' "${array[@]}" | grep 'match'
```

integers

```
(( $1 == 1 || $1 == 2 )) && echo "number 1 or 2"
((i-=2)) # decrements $i by 2
i=0; echo $((i+=1))
n=2; if ! (( $n == 1 )); then echo 'not 1'; fi
n=1; printf "%03d\n" $n
n=08; (( 10#$n > 7 )) && o base10 # because 08 is an impossible octal
```

comparison

```
if [ "$a" -ge 2 ]
if [ "$a" -gt 2 ]
if [ "$a" -le 2 ]
if [ "$a" -lt 2 ]
if [ "$a" -ne 2 ]
```

((>))

```
(( 1 > 0 )) && o 1
if (( 1 > 0 )); then echo greater; fi
if (( 1 >= 0 )); then echo greater; fi
```

empty or undefined

```
$undefined && echo 'Eh! $undefined true!'
empty=''; [ -z $empty ] && echo empty
```

strings

```
[[ $t =~ (y|n) ]] && echo 'good answer'
[ 'y' == 'n' ] || echo 'nope'
n=''; n=n; [[ -n $n ]] && echo $n
o lkj | rev | cut -c 2- | rev
o ljk | sed 's/.$//'
s='long'; o ${#s} # length
qs='$quote\*star'; o "$qs"
s=12345; echo $s | awk '{print substr($1,length($1)-2) }'
```

```
s=12345; echo $s | cut -c $(( ${#s}-2 ))-
s=yes; s+=no; o $s
s=start; o $s'-end'
s+=new; o $s
${x}and$y
```

case conversions: var=vAlUe; o \${var^^}; o "\${var,,}"

\$PATH

```
echo "${PATH//:/$'\n'}"
echo "$PATH" | tr ':' '\n'
```

script's name

```
DIR=$(dirname "${BASH_SOURCE[0]}") # get the directory name
DIR=$(realpath "${DIR}") # resolve its full path if need be
```

parameter expansion

```
file=a.b.c; echo ${file##*.}; echo ${file%.*}
```

```
s='strong string'; o ${s//str/p} # all matches
${string/substring/firstMatchReplacement}
${string/#substringAtStart/replacement}
${string/%substringAtEnd/replacement}
${string:position:length}
a=12345; o ${a:0:2}; o ${a:2}; o ${a:1:2}; o ${a:2:1}
b="a(b(c(d"; o ${b##*\()}
s=12345; echo ${s::-2}; echo ${s: -2}
t=lkj; echo ${t:0:${#t}-1}
```

```
name=polish.ostrich.racing.champion; o ${name##*.}; o ${name###*.}; o ${name%%.*}; o ${name%*.*}
u=.git.git.git; o ${u%.git}
```

flow control

- ctrl+q continue
- ctrl+s stop

sort

```
sort -o <file> <file> # in-place
```

- -h (--human-numeric-sort)
- -n (--numeric-sort)
- -r (--reverse)
- -o (--output=FILE)
- -u (--unique)
- -V (--version-sort) natural sort of (version) numbers within text
- --help
- --version

system

```
cat /etc/hostname
clamscan -r # shows Loading & Compiling
clamscan -r 2>&1 | tee clamscan.log
gtk-launch --version
```

```

halt -p
i hier # detailed description of the filesystem hierarchy
locale
openbox --reconfigure
passwd jo # then re-login
ps $(pgrep Xorg) # shows which tty X is on
shutdown now
swapon --show
w # list users and load on system
whereis <executable>
xrdb -query -all # shows loaded X resources
xset q # shows a variety of IO settings

```

- awesomewm: floating window: winkey+left_mouse_drag
- lsmod(8) show what kernel modules are currently loaded
- maximum 255 bytes per filename & 4096 per path

awesome wm

- maximized (horizontally or vertically) are indicated by (horizontal or vertical) double arrow, and break tiling -modkey+left_mouse_drag move -modkey+right_mouse_drag resize -modkey+s awful.hotkeys_popup
- on top clients are indicated by a caret

boot

```

bootctl -h # --help
find /dev/disk/by-designator -type l -ls # recursively list all symlinks with their referen

```

conky

```

conky -C # --print-config
killall -SIGUSR1 conky # brings it back up

```

CPU-X

```

cpu-x -h
cpu-x -N
Openbox Menu > System > CPU-X

```

directory sizes as root

```

du -shx
ncdu -x # --one-file-system

```

etckeeper

```

/etc/.gitignore
/etc/etckeeper/etckeeper.conf
sudo ls /etc/.git

```

groups

```

/etc/group # all of them
gpasswd [-a <user> <group_to_add_to>]
groups jo
sudo groupadd <group_to_create>
sudo groupdel <group_to_delete>

```

GRUB

```
grub-install --version
```

(GRand Unified Bootloader)

kernel parameters

```
sysctl -a # display all kernel parameters
sysctl -A # list all parameters
sysctl -T # list parameters in a tree
```

lnav

```
journalctl -f | lnav
journalctl -f -o json --output-fields=MESSAGE,PRIORITY,_PID,SYSLOG_IDENTIFIER,_SYSTEMD_UN
journalctl -f -o json | lnav
journalctl -f -o short-iso | lnav # handling year changes
```

- hit Esc to get out of weird edit mode
- The Logfile Navigator

monitoring

```
atop
free -h
glances
htop
iostat -dm -p sda2 3
last reboot
sair -c
```

dool

```
dool --help
dool -dD sda # --disk
dool -n # --net
dool -s # --swap
i dool
```

notification

```
zenity --calendar
```

Dunst

```
dunstctl history-pop # repeat for previous messages
dunstify --help
dunstify -u critical "Read this now!"
dunstify "First test message."; dunstify "Second test message, which is longer."
dunstrc allows tweaks that aren't available with Xfce Notify Daemon
```

dunst

```
dunst &
pgrep dunst
pkill dunst
```


- (if `xfce4-notifyd` hasn't taken over `org.freedesktop.Notifications`) any notification reloads `dunst`
- takes over `org.freedesktop.Notifications` thus incapacitating `xfce4-notifyd`
- only keeps `history_length` of notifications in memory, nothing persistent

close notifications

```
dunstctl close # the last one
dunstctl close-all
```

if no luck, can also kill the `Xfce Notify Daemon`

notify-send

```
notify-send 'test of notify-send'
notify-send --help
notify-send -u critical "test of critical notification"
notify-send -t 5000 -u low "low urgency 5s notification"
```

Plasma

```
Ctrl+Alt+Shift+Del -> exit KDE without saving
Ctrl+Alt+Del       -> logout screen
Ctrl+Alt+T         -> Konsole
Win+PageUp/Down   -> maximises or returns to normal a window
```

Qt

```
qmake -query QT_VERSION
qmake -v also reports Qt version
```

systemd

```
systemctl halt
systemctl reboot
systemctl suspend
systemd-analyze --system unit-paths
systemd-analyze --user unit-paths
systemd-analyze calendar "*-*-* *:00:00"
```

info

```
systemctl --failed
systemctl --user --no-pager # list running Systemd units
systemctl --version
systemctl is-enabled <service>
systemctl list-timers
systemctl list-unit-files --no-pager
systemd-analyze blame # time taken for boot processes
```

journalctl

```
:Man journalctl
journalctl --disk-usage
journalctl --list-boots
journalctl --verify
```

- `-f`, `--follow`
- `-n`, `--lines=` (default 10) most recent events
- `-u` (`--unit=UNIT|PATTERN`)

- -r (= --reverse) output
- journalctl(1)

latest

- journalctl -f | lazyjournal I don't get...
- with lnav

messages, paged

```
journalctl -b # for this boot
journalctl -b | tl # (toolong) a little slow, but easier to read
journalctl -b -1 # for previous boot
journalctl -b -1 -e # shows end of -n1000
journalctl -b -1 -r # newest first
journalctl -b -g 'EFI v' # shows EFI version
```

uname

uname -a (= --all) handy line of system info

--kernel-release

```
[[ $(uname -r) =~ 'arch' ]] && echo 'Arch Linux'
[[ $(uname -r) =~ 'microsoft' ]] && echo 'WSL2'
```

users

```
[[ $EUID > 0 ]] && echo "not root"
id -u
su -
who # lists users active on terminals
whoami
```

Xfce

```
xfce4-display-settings &
xfce4-panel -r # reloads
xfce4-settings-manager &
```

- GTK
- Terminal (xfce4-terminal)
 - ctrl+shift+c = copy
 - ctrl+shift+v = paste
- xfwm4-tweaks-settings & = Windows Manager Tweaks

Clipman

- ~/.cache/xfce4/clipman/textsrc semicolon-separated list
- clickable Panel icon

keyboard shortcuts

- Alt+F3 = Application Finder
- Alt+scrollwheel = zoom_desktop
- Ctrl+Alt+D = minimize all
- Ctrl+Alt+Del = xfce4-session-logout
- Ctrl+Alt+Escape = xkill (right-click abandons)
- Ctrl+Alt+L = xflock4
- HomePage = XF86HomePage = exo-open --launch WebBrowser

- Shift+Ctrl+Esc = xfce4-taskmanager
- Super+e = thunar
- Applications > Settings > Keyboard > Application Shortcuts can take a bash script path

windows

- Applications > Settings > Window Manager
 - Alt+F9 = minimise
 - Alt+F10 = toggle maximise
 - F11 = full screen
 - super+1 = move to left monitor
 - super+2 = move to right monitor

workspaces

- Ctrl+Alt+Home/End = move window to left/right workspace
- Ctrl+Alt+left/right arrow = left/right workspace
- Ctrl+Fn = goto workspace n

Thunar

~/.config/xfce4/xfconf/xfce-perchannel-xml/thunar.xml

Custom Actions: ~/.config/Thunar/uca.xml

keyboard accelerators

- ~/.config/Thunar/accefs.scm accelerator map dump
- Ctrl+Shift+n = create-folder

Xfce Notify Daemon

```
systemctl --user status xfce4-notifyd
systemctl --user start xfce4-notifyd
systemctl --user stop xfce4-notifyd
xfce4-notifyd-config &
```

no indication of priority - only distinguishing feature of critical notifications is their persistence

SystemRescue

```
setkmap fr
startx
```

term

```
sudo fgconsole # reports tty number
```

- ^\ = SIGQUIT
- ^c = SIGINT
- ^q resume output to screen
- ^s pause output to screen
- w(1) logged on info

Alacritty

```
alacritty -h # --help
alacritty -V # --version
```

- Ctrl+Shift+b = SearchBackward, then Esc
- Ctrl+Shift+space = ToggleViMode

font

- Ctrl+- = DecreaseFontSize
- Ctrl+0 = ResetFontSize
- Ctrl+= = IncreaseFontSize

scroll

- Shift+PageDown/Up = ScrollPageDown/Up
- Shift+End/Home = ScrollToBottom/Top

size

```
echo $COLUMNS $LINES
stty size # lines columns
tput cols lines
```

set (not in tmux)

```
printf '\033[8;40;100t'
resize -s 90 150
```

terminal line settings

```
stty -a
```

flow control

```
[-]ixon = the setting
ctrl-q = XON
ctrl-s = XOFF
```

Termux

```
if ! [ $TERMUX_VERSION ]; then echo 'not in Termux'; fi
termux-info
```

tmux

```
if [ $TERM == 'screen-256color' ]; then echo "you're in tmux"; fi
pgrep tmux -l
```

\$ tmux ...

```
tmux a                # attach
tmux a -t myname       # attach to session myname
tmux detach
tmux kill-server       # good for resetting
tmux kill-session -a   # kills all but current session
tmux list-keys | grep '~'
tmux list-keys | mo
tmux ls                # list sessions
tmux lsk -Na           # includes my rebinds
tmux send ls enter     # (send-keys) ls in the currently active pane
tmux show -g prefix    # (show-options)
tmux show -s           # server
```

internal commands

```
:bind # alias for bind-key
:choose-tree # put a pane into tree mode, can kill and select
:set synchronize-panes # (set-option)
```

join-pane

```
:j-s 2.1 [-t 1.0 ] # src-pane W2P2 [to dst-pane W1P0]
```

- :joinp is the only abbreviation I can find in the manual
- from another pane, :j-hs 2.1 pulls in src-pane W2P2 alongside

tput

```
(x=`tput op` y=`printf %76s`;for i in {0..256};do o=00$i;echo -e ${o:${#o}-3:3} `tput setaf $i;tput setab $i`${y// /=}$x;done) # 256 lines of colour
tput bold; tput setaf 3; tput setab 4; echo 'bold yellow on blue'
tput colors
tput cuu 2; tput el # move back a line and clear it
```

urxvt

```
urxvt --help
```

- \$TERM is set to rxvt-unicode-256color
- rxvt-unicode
- URxvt.url-launcher

tr

```
tr '\n' '~' < <in> | sed 's/~ / /g' [> <out>] # replace newlines with double spaces
tr (Unix)
```

Ubuntu

```
lsb_release -a
ods2tsv list.ods > list.tsv
sensible-browser https://manpages.ubuntu.com/manpages/jammy/man1/sensible-browser.1.html
sudo updatedb

sed -i will change fileformat to dos
```

packages

```
apt-cache search .
```

```
https://packages.ubuntu.com/
```

manage

```
apt-cache -h
apt-cache show <package> # info
apt-cache showpkg <package> # report includes dependencies and reverse dependencies
apt-get download <package> # to current directory
dpkg -l
dpkg-deb -x <package> <directory> # --extract
sudo apt install <package1> <package2> ...
sudo apt remove <package1> <package2> ...
```

- sudo apt update [the package index]
- sudo ls /etc/apt/sources.list.d/ repositories

upgrade packages

```
sudo apt autoremove
sudo apt upgrade
sudo apt-get clean
sudo apt-get dist-upgrade # better, riskier
```

Pro

```
pro security-status
pro status
sudo pro attach <FPT>
```

vim

```
$HOME/.viminfo
e -la /usr/bin/vim
find . -type f -name tags
file "$(command -v vim)"
gvim &
neovide &
```

WAN

```
if wget -q --spider google.com; then echo online; fi
pkill radio; radio -K # quit & kill instances of radio-active
```

Carburetor

```
flatpak info io.frama.tractor.carburetor
```

commands

- Ctrl+, = Preferences
- Ctrl+? = Shortcuts
- Main Menu > Set Proxy then <https://check.torproject.org/>

Chawan

```
cha -V
cha https://en.wikipedia.org
mancha cha
```

key bindings

- alt+i toggle images
- q quits
- v toggles select
- y yank to clipboard
- vim-like

email - clm

```
$OSL/nodes/bashrc-clm
mbsync -v
```

msmtp

```
msmtp --version
```

"an SMTP client"

neomutt

```
neomutt -D # dump settings
neomutt -h # help
neomutt -v # version
rm -r ~/.cache/mutt/*
sidebar_format
```

notmuch

```
notmuch config get database.path
```

search

```
nmse ...
notmuch --help
notmuch --version
notmuch search '"the Pennines"' # finds exactly that
notmuch search tag:attachment | wc -l
notmuch search tag:fm | wc -l
notmuch search tag:zou | wc -l
notmuch search tag:zou and tag:inbox | wc -l
notmuch tag -cz -- tag:zou # removes all cz tags
```

- case-insensitive
- date:DD-MM[-[YY]YY]
- date:D[D].M[M][.[YY]YY]
- date:YYYY-MM[-DD]
- from:
- NOTMUCH-SEARCH-TERMS(7)
- seems to not find emails that're included in subsequent ones
- subject:(pizza free) (= subject:pizza and subject:free) matching "Free Delicious Pizza" (while subject:"pizza free" won't)
- to:

email - Thunderbird

```
thunderbird -addressbook
thunderbird -h ## --help
```

version

```
thunderbird --full-version
thunderbird -v ## --version
```

firewall

```
sudo iptables -L
```

Firewalld

```
sudo firewall-config
sudo systemctl start firewalld
sudo systemctl stop firewalld
sudo systemctl status firewalld
```

```
sudo systemctl restart firewallld
```

firewall-cmd

```
sudo firewall-cmd --info-zone=home
sudo firewall-cmd --info-zone=public
sudo firewall-cmd --panic-on
sudo firewall-cmd --state
```

iptables

```
sudo iptables -nvL
sudo systemctl status iptables.service
```

GNU Wget

:Man wget

```
wget -kr -A.zip https://url-to-webpage-with-pdfs/ # works from that page
```

```
wget -r -A.pdf http://url-to-webpage-with-pdfs/ # works recursively from root page
```

- -E (--adjust-extension)
- -O file (--output-document=file)

rdrview

i rdrview

```
rdrview http://harriott.github.io -M
```

transmission-cli

```
http://localhost:9091/transmission/web/
```

```
transmission-daemon --dump-settings
```

```
transmission-remote -a some.iso.torrent
```

```
transmission-remote -l
```

w3m

```
w3m http://harriott.github.io # showing images
```

```
w3m https://man.archlinux.org/man/w3m.1.en
```

keymaps

- H Current keymap file
- M (= EXTERN) opens page in default browser
- o (= OPTIONS) Option Setting Panel
 1. [shift-]tab to option then Enter
 2. tab to [OK] then Enter to end
- Q (= EXIT)
- r (= VERSION)
- R (= RELOAD)

movements

- + (= NEXT_PAGE)
- - b (= PREV_PAGE)
- [,] (= LINK_BEGIN, LINK_END) start, end of page
- g (= BEGIN) start of page
- w (= NEXT_WORD)
- W (= PREV_WORD)
- z (= CENTER_V) centres on current line
- vim-like

navigation

B (= BACK)

URLs

- c (= PEEK) reveals the url
- e (= PEEK_IMG) reveals the url
- U (= GOTO) presents dialogue on current url (ctrl-u clears that url)
- u (= PEEK_LINK)
- v (= VIEW) toggles HTML view

page contents

- I (= VIEW_IMAGE)
- S (= SAVE_SCREEN) requests name for file to save to (line breaks are kept)
- / (= SEARCH) centres on found
- tab, shift+tab jump to next, previous link

tabs

- ctrl+q (= CLOSE_TAB)
- s (= SELECT_MENU) to move tabs
- T (= NEW_TAB) re-opens current tab, new
- { (= PREV_TAB)
- } (= NEXT_TAB)